7 The Physics of High-Performance Transit: Singapore's R151 MRT Trains

The Singapore Mass Rapid Transit (MRT) system recently introduced the seventh-generation Alstom Movia R151 rolling stock for the North–South and East–West lines. These trains operate using a 750 V d.c. third-rail electrical supply.

Electrical power from the national grid is supplied to traction substations located along the MRT line. At the Buona Vista (BNV) substation, electrical power is received from the national grid at an alternating potential difference of 22 kV. A step-down transformer reduces this to a r.m.s. potential difference of 750 V a.c., before a rectifier converts the a.c. supply to d.c. for the third rail. The transformer can provide a r.m.s. secondary current of 7200 A.

A single trainset consists of 6 cars in a DT–M1–M2 + M2–M1–DT formation, where 'M' denotes a motored car and 'DT' denotes a trailer car with a driver's desk. To enhance efficiency, these trains are constructed from lightweight aluminium alloy, giving a 6-car set a length of 138.86 m and a width of 3.2 m. The car height is approximately 3.7 m. A fully loaded train has a mass of approximately 300 tonnes where 1 tonne equals to 1.0×10^3 kg.

While designed for a maximum speed of 90 km h^{-1}, the trains typically operate at a service speed of 80 km h^{-1}. To ensure safety, automated systems maintain a constant deceleration of 1.3 m s^{-2} during emergency stops.

Sustainability is a core focus for the Land Transport Authority (LTA). R151 trains utilize high-efficiency traction systems capable of regenerative braking. In this mode, the traction motors act as generators, converting the train's kinetic energy into electrical energy that is fed back into the third-rail for use by other trains on the line. The effectiveness of this recovery depends on the line receptivity—the requirement that other trains must be operating nearby to immediately consume the electricity as it is generated. If no other trains are in the vicinity to use the power, the system is 'non-receptive,' and the excess energy will be dissipated as heat through on-board resistors.

A train in motion experiences total resistance consisting of mechanical friction and air drag. Generally, the resistance W_0 (in newtons) is often modeled by the Davis equation:

$$W_0 = a + bv + cv^2$$

where a, b and c are empirical coefficients and v is the velocity.

For a 300-tonne SMRT train, the specific resistance R (in newton per tonne) in open air is defined as:

$$R = 2.3 + (0.6v^2 \times 10^{-3})$$

where v is the speed in km h^{-1}.

In confined underground tunnels, air resistance is significantly amplified by the "piston effect". As a train moves, it compresses the air column in front of it while creating a low-pressure wake behind it. The magnitude of this effect is governed by the blockage ratio (β), the ratio of the train's frontal area (A_{tr}) to the tunnel's cross sectional area (A_t):

$$\beta = \frac{A_{tr}}{A_t}$$

Research into the high-blockage tunnels has identified a critical transient phenomenon during the deceleration phase:

1. Steady-state motion: At constant velocity, the air resists the train's motion (negative drag), requiring motor power to overcome.
2. Transient deceleration: As the train brakes, the air column trapped behind it continues to move forward due to its own momentum. This air column exerts a positive force on the rear of the train, essentially "pushing" it in the direction of travel.

In the research, the variation of train drag coefficient C_D with time t of scaled model trains with different β values in an experiment is shown in Fig. 7.1.

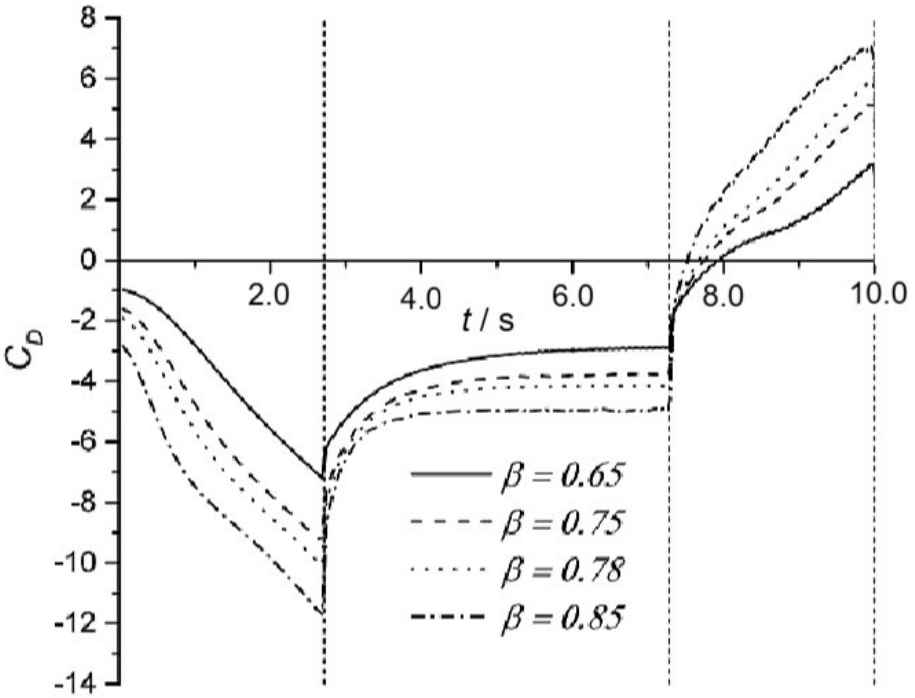


Fig. 7.1

During the experiment, the train first accelerates at 1.0 m s^{-2}, then moves with constant speed and then decelerates in the confinement of a tunnel wall.

The drag coefficient C_D is a dimensionless quantity that characterizes the aerodynamic resistance of an object based on its shape and the fluid environment. In this experiment result, a negative drag coefficient C_D means that the fluid pushes the object backwards while a positive drag coefficient C_D means the fluid pushes the object forward.

(a) Determine the S.I. base units of the coefficient c in the Davis equation.

base units of c = ... [2]

(b) **(i)** For a SMRT train cruising at service speed in open air, determine the resistance acting on the train.

resistance = ... N [2]

(ii) In a confined tunnel, the resistance acting on the train cruising at service speed is increased to 42.5 kN due to the "piston effect".

1. Determine the output power of the train to overcome the resistance force.

output power = ... W [2]

2. Given that the train draws a current of 2100 A, determine the efficiency of the train.

efficiency = ... % [2]

3. Suggest why the efficiency of the train is not 100%.

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.....[1]

(c) For the Buona Vista substation,

(i) calculate the turns ratio (secondary turns to primary turns) for the transformer used in this substation,

turns ratio = ... [2]

(ii) calculate the peak current drawn by the substation from the 22 kV grid, assuming the transformer operates with 100% efficiency.

peak current = ... A [2]

(d) **(i)** If the regenerative system captures 30% of kinetic energy of the train, determine the amount of electrical energy produced through the regenerative braking system when a fully loaded train enters a station.

electrical energy produced = ... J [2]

(ii) Official LTA data states that regenerative braking allows trains to recuperate approximately 30% of the total energy consumed.

Explain why the percentage might decrease during off-peak hours when fewer trains are running on the line.

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.....[1]

(e) **(i)** Determine the cross-sectional area of the tunnel if the SMRT wants to design the underground track system to have a blockage ratio of 0.65.

area = ... m^2 [1]

(ii) In Fig. 7.1, the experimental data shows that the drag coefficient becomes positive at approximately $t = 7.5$ s.

Explain the physical cause of this "positive force" and its impact on the braking system.

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.....[2]

(iii) A researcher claims that increasing the blockage ratio has the most severe impact on the train's braking systems rather than its acceleration systems.

The maximum and minimum values of C_D due to different blockage ratio (β) is shown in Fig. 7.2.

	Max C_D value	Min C_D value	Percentage increase
Peak resistive drag during acceleration	11.8	7.2	64%
Peak forward push during deceleration			

Fig. 7.2

Complete Fig. 7.2 with data from Fig. 7.1 and evaluate the researcher's claim.

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[Total: 21]